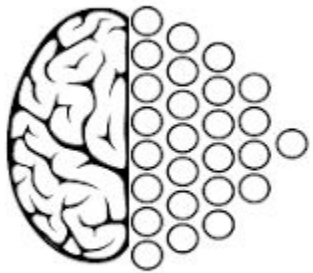
ML Without Labels

SAIDL x CTE (Lecture 9)



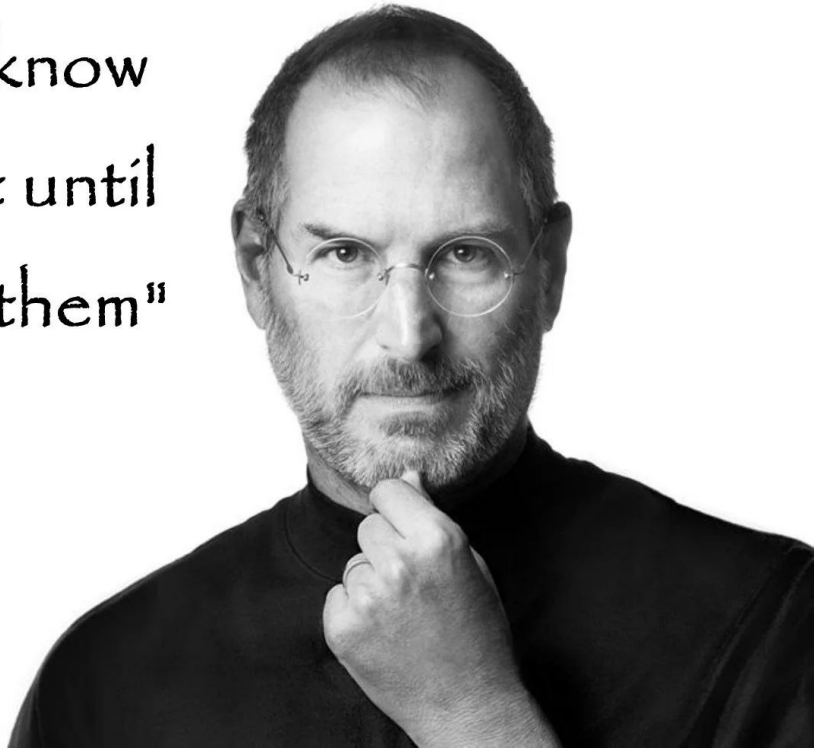
Journey so far - U tell me what it is (E**x**’s and Wh**y**’s)

~~Unit 1: Foundations~~	~~L1~~	~~Overview~~	~~What is ML?; real-life applications; linear and logistic regression.~~
~~Unit 1: Foundations~~	~~L2~~	~~How ML Models Learn~~	~~Supervised learning; loss functions; gradient descent; classifiers.~~
~~Unit 1: Foundations~~	~~L3~~	~~Building a Neural Network~~	~~Neural networks; backpropagation; activation functions.~~
~~Unit 1: Foundations~~	~~L4~~	~~Making Neural Networks Learn~~	~~Optimization; dropout; initialization; regularization.~~
~~Unit 2: Vision and Sequences~~	~~L5~~	~~Computer Vision~~	~~Convolutional neural networks; pooling; feature maps.~~
~~Unit 2: Vision and Sequences~~	~~L6~~	~~Sequential Models~~	~~ResNets; recurrent neural networks; long short-term memory networks.~~
~~Unit 3: Attention and Transformers~~	~~L7~~	~~Attention~~	~~Attention; queries, keys, and values; self-attention; cross-attention; tokenization.~~
~~Unit 3: Attention and Transformers~~	~~L8~~	~~Transformer Architecture~~	~~Multi-head attention; positional encoding; the Transformer.~~

But here is the catch !!!!

"People don't know what they want until you show it to them"

~Steve Jobs



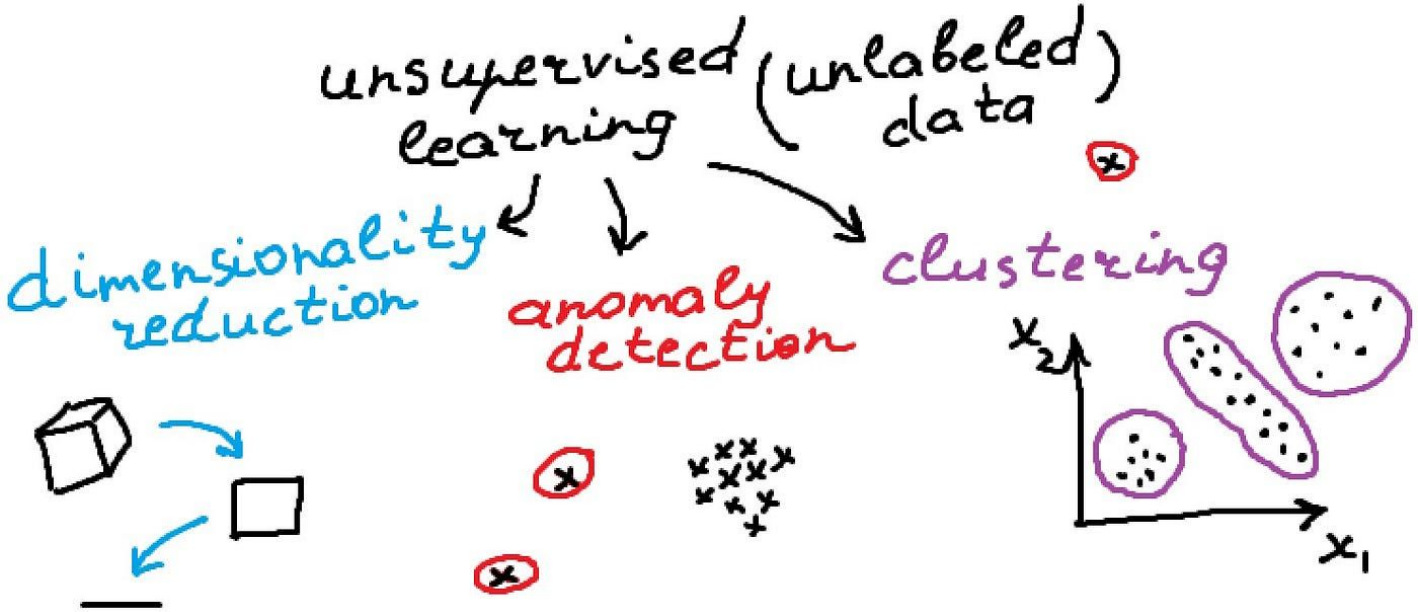
CORE PROBLEM - MOVING TO UNSUPERVISED SETUP

Real world doesn’t come with supervised setup.

I.e nicely defined X and Y; where all u do if fitting the lines.

UNSUPERVISED LEARNING

U are given with X and Goal is to **find some kind of structure in it**.



K- means clustering

In the clustering problem, we are given a training set $\{x^{(1)}, \ldots, x^{(n)}\}$, and want to group the data into a few cohesive "clusters." Here, $x^{(i)} \in \mathbb{R}^d$ as usual; but no labels $y^{(i)}$ are given. So, this is an unsupervised learning problem.

The k-means clustering algorithm is as follows:

1. Initialize **cluster centroids** $\mu_1, \mu_2, \ldots, \mu_k \in \mathbb{R}^d$ randomly.

2. Repeat until convergence: {

   For every i, set

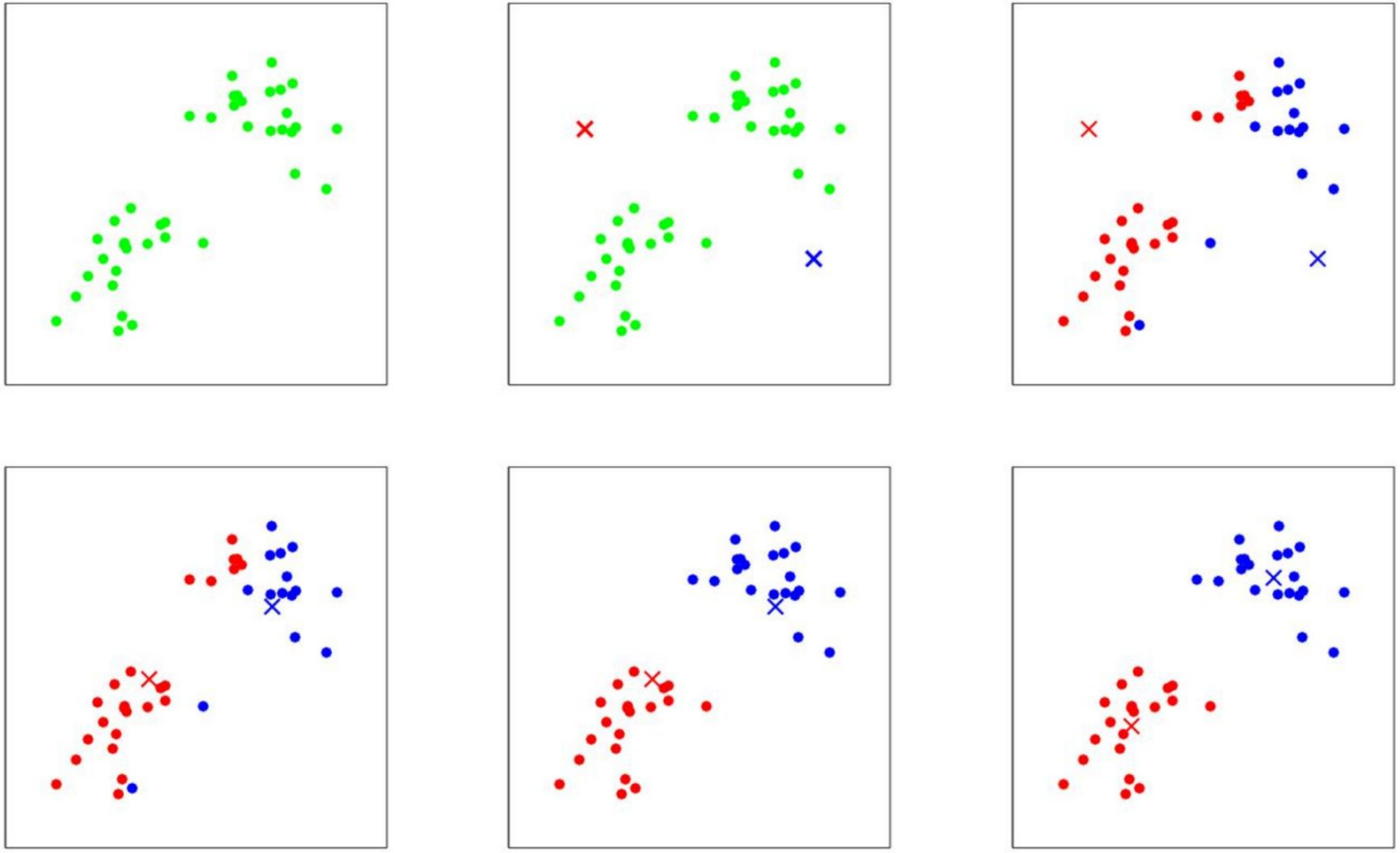
   $$c^{(i)} := \arg\min_j ||x^{(i)} - \mu_j||^2.$$

   For each j, set

   $$\mu_j := \frac{\sum_{i=1}^{n} 1\{c^{(i)} = j\} x^{(i)}}{\sum_{i=1}^{n} 1\{c^{(i)} = j\}}.$$

   }

TL;DR
1 - find cluster for each point
2 - update centroid ; 3 - repeat



But what about convergence?

Is the k-means algorithm guaranteed to converge?

Yes it is, in a certain sense. In particular, let us define the distortion function to be:

$$J(c, \mu) = \sum_{i=1}^{n} ||x^{(i)} - \mu_{c^{(i)}}||^2$$

The key insight is that K-means is a **form of coordinate descent, also called alternating minimization**.

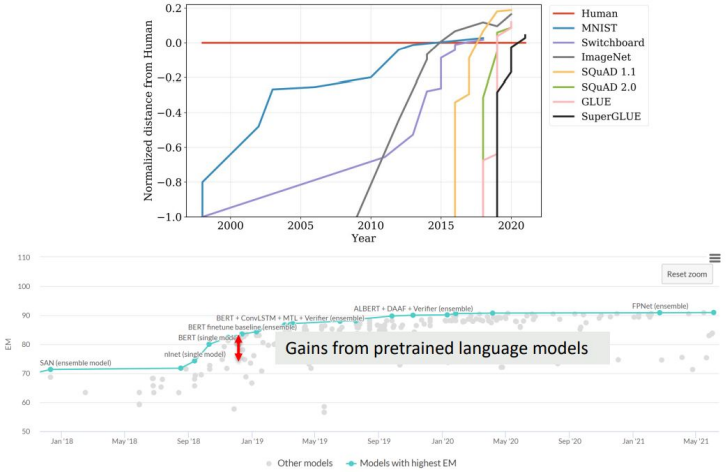
The distortion function J is a **non-convex function**, and so coordinate descent on J is not guaranteed to converge to the **global minimum**. In other words, **k-means can be susceptible to local optima**

If you are worried about getting stuck in bad local minima, one common thing to do is run k-means many times.

Question - Why things can u change in different runs? And how do u pick which one is the best ?

How Modern AI deals without Labels

The pretraining revolution

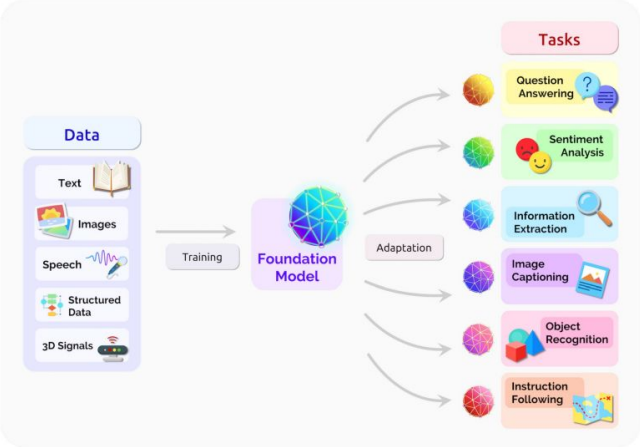


Pretraining has had a major, tangible impact on how well NLP systems work

* Thanks to CS224n for slides and material

Simple idea - scale things up with data and compute and u are good (Well Maybe not anymore; wait for it)

Pretraining – scaling unsupervised learning on the internet



Key ideas in pretraining

- Make sure your model can process large-scale, diverse datasets
- Don’t use labeled data (otherwise you can’t scale!)
- Compute-aware scaling

Language Modelling

Word structure and subword models

Let's take a look at the assumptions we've made about a language's vocabulary.

We assume a fixed vocab of tens of thousands of words, built from the training set.
All *novel* words seen at test time are mapped to a single UNK.

	word		vocab mapping	embedding
Common words	hat	→	pizza (index)	
	learn	→	tasty (index)	
Variations	taaaaasty	→	UNK (index)	
misspellings	laern	→	UNK (index)	
novel items	Transformerify	→	UNK (index)	

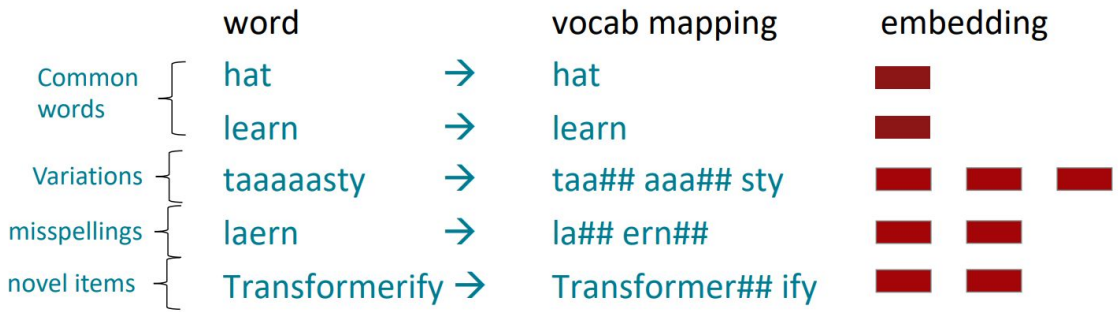


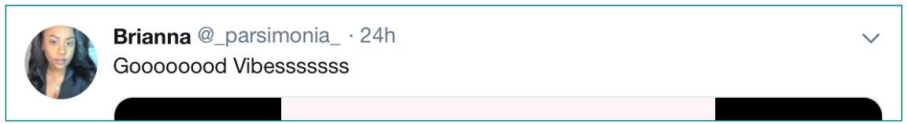
Can we do better maybe ?? (Bonus if u remember what algorithm is used for this, recall tokenization)

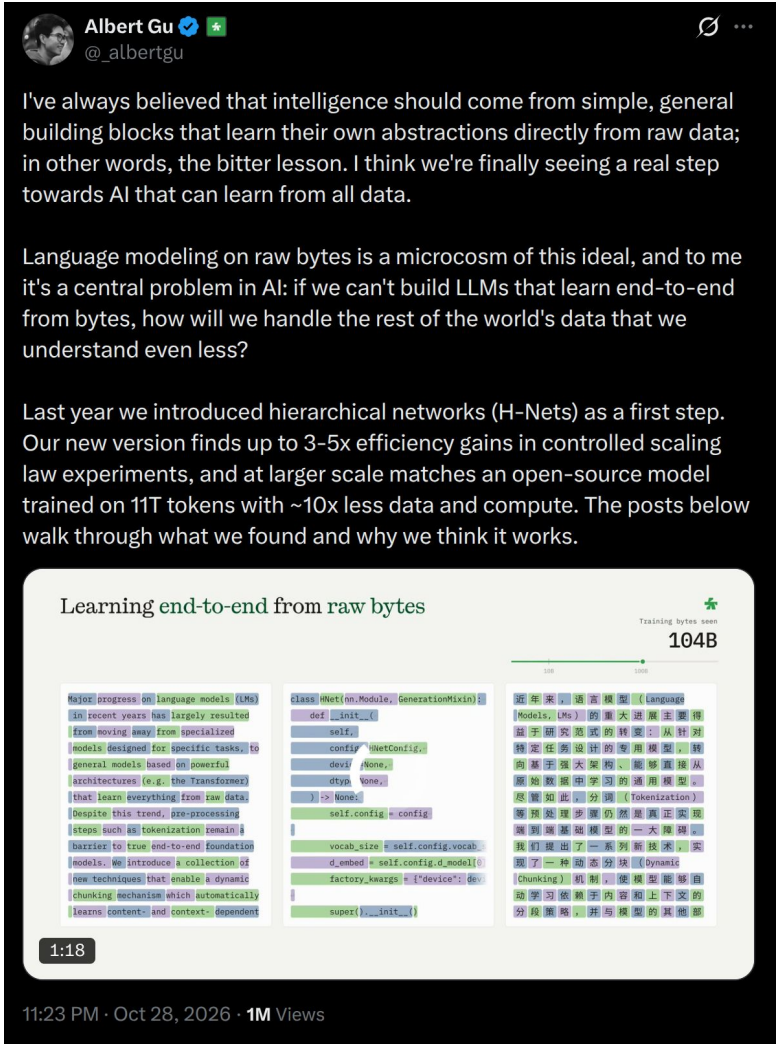
Word structure and subword models

Common words end up being a part of the subword vocabulary, while rarer words are split into (sometimes intuitive, sometimes not) components.

In the worst case, words are split into as many subwords as they have characters.







Can we get rid of tokenizer?

Check this out :- H-nets

Motivating word meaning and context

Recall the adage we mentioned at the beginning of the course:

> *"You shall know a word by the company it keeps"* (J. R. Firth 1957: 11)

This quote is a summary of **distributional semantics**, and motivated **word2vec**. But:

> "… the complete meaning of a word is always contextual,
> and no study of meaning apart from a complete context
> can be taken seriously." (J. R. Firth 1935)

Consider *I* ***record*** *the* ***record***: the two instances of ***record*** mean different things.

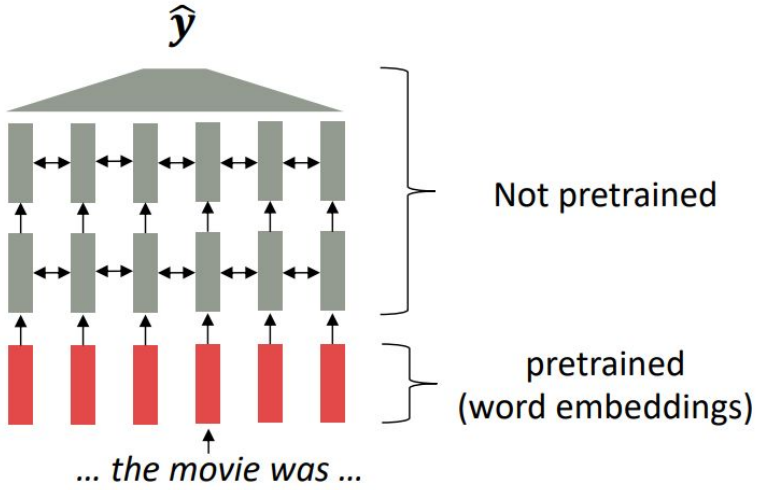
Where we were: **pretrained word embeddings**

Circa 2015:

- Start with pretrained word embeddings (no context!)
- Learn how to incorporate context in an LSTM or Transformer while training on the task.

Some issues to think about:

- The training data we have for our **downstream task** (like question answering) must be sufficient to teach all contextual aspects of language.
- Most of the parameters in our network are randomly initialized!



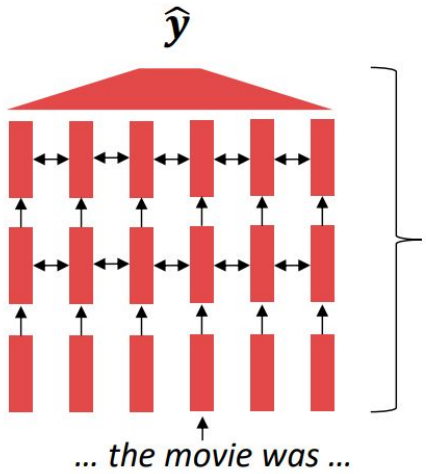
[Recall, *movie* gets the same word embedding, no matter what sentence it shows up in]

Where we're going: **pretraining whole models**

In modern NLP:

- All (or almost all) parameters in NLP networks are initialized via **pretraining**.
- Pretraining methods hide parts of the input from the model, and train the model to reconstruct those parts.

- This has been exceptionally effective at building strong:
  - **representations of language**
  - **parameter initializations** for strong NLP models.
  - **Probability distributions** over language that we can sample from



[This model has learned how to represent entire sentences through pretraining]

What can we learn from reconstructing the input?

It's just FILL IN THE BLANKS !!!

Stanford University is located in _____, California.

I put ___ fork down on the table.

Overall, the value I got from the two hours watching
it was the sum total of the popcorn and the drink.
The movie was ___.

I was thinking about the sequence that goes
1, 1, 2, 3, 5, 8, 13, 21, ____

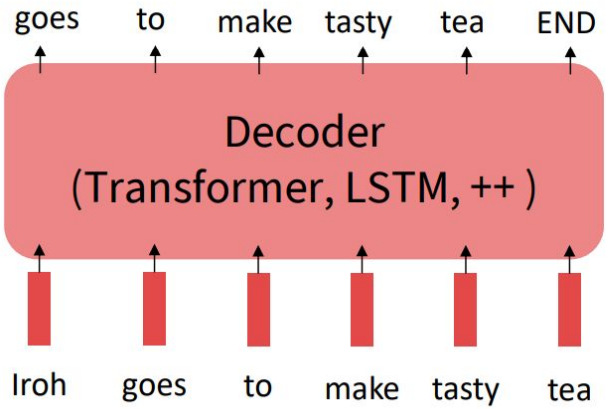
Pretraining through language modeling [Dai and Le, 2015]

Recall the **language modeling** task:

- Model $p_\theta(w_t|w_{1:t-1})$, the probability distribution over words given their past contexts.
- There's lots of data for this! (In English.)

Pretraining through language modeling:

- Train a neural network to perform language modeling on a large amount of text.
- Save the network parameters.

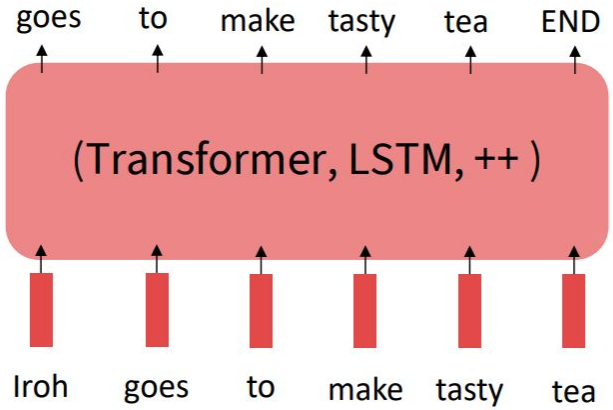


The Pretraining / Finetuning Paradigm

Pretraining can improve NLP applications by serving as parameter initialization.

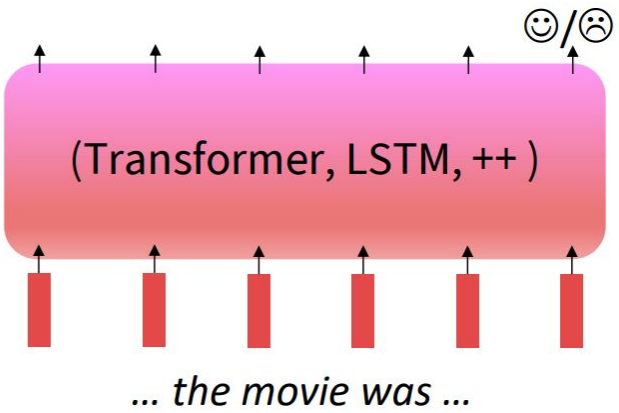
Step 1: Pretrain (on language modeling)

Lots of text; learn general things!

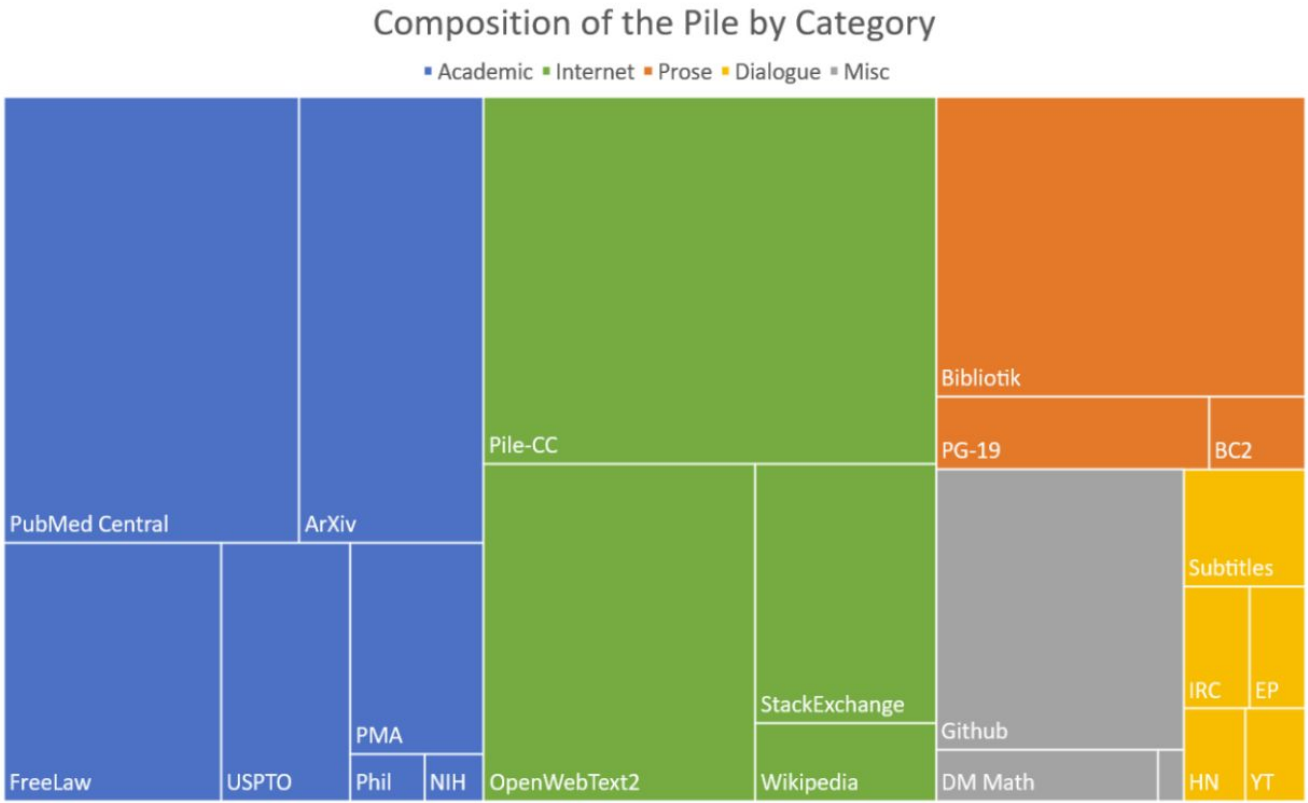


Step 2: Finetune (on your task)

Not many labels; adapt to the task!

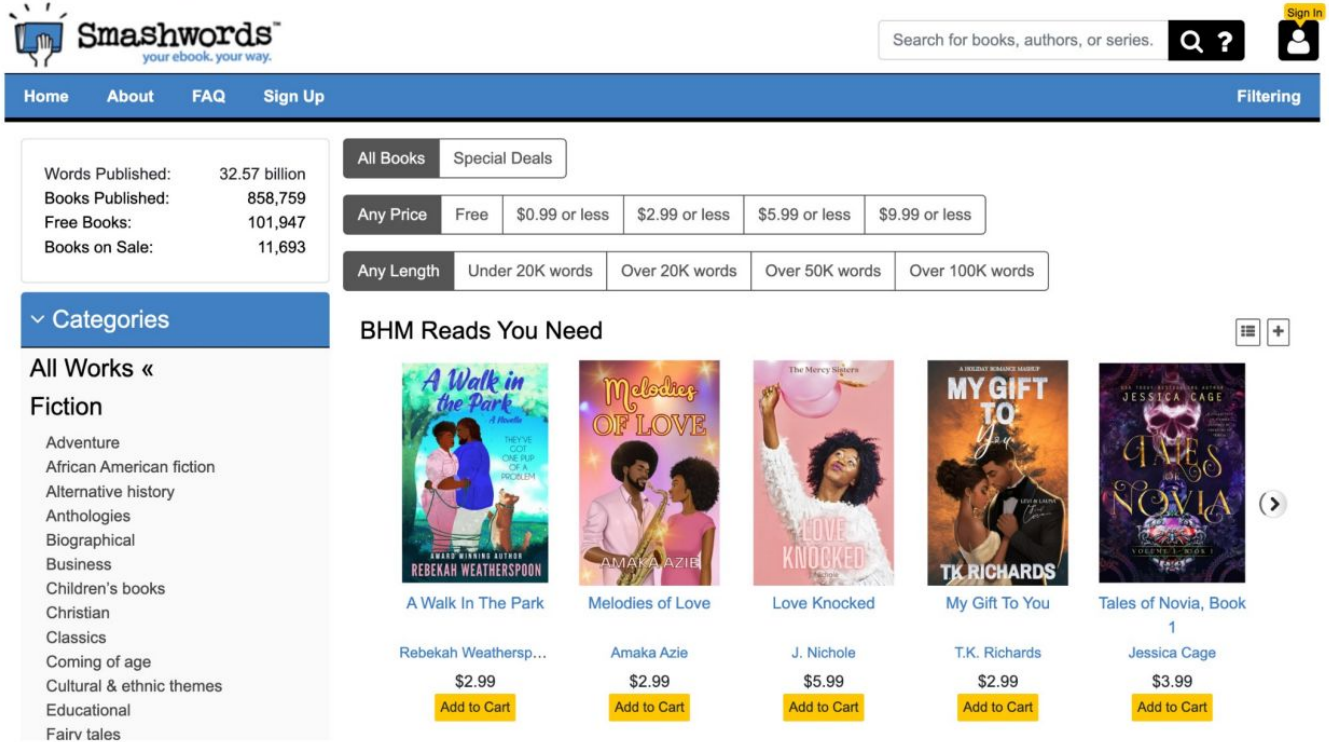


Where does this data come from?



Model	Training Data
BERT	BookCorpus, English Wikipedia
GPT-1	BookCorpus
GPT-3	CommonCrawl, WebText, English Wikipedia, and 2 book databases (“Books 1” and “Books 2”)
GPT-3.5+	Undisclosed

Bookcorpus … what's that?

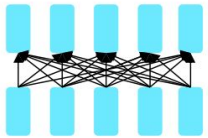


- Scraped ebooks from the internet – highly controversial

NEXT LECTURE ????

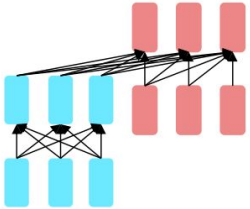
Pretraining for three types of architectures

The neural architecture influences the type of pretraining, and natural use cases.



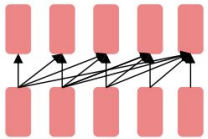
Encoders

- Gets bidirectional context – can condition on future!
- How do we train them to build strong representations?



Encoder-Decoders

- Good parts of decoders and encoders?
- What's the best way to pretrain them?



Decoders

- Language models! What we've seen so far.
- Nice to generate from; can't condition on future words

What kinds of things does pretraining teach?

There's increasing evidence that pretrained models learn a wide variety of things about the statistical properties of language. Taking our examples from the start of class:

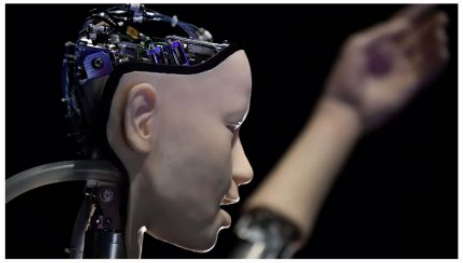
- *Stanford University is located in _____, California.* [Trivia]
- *I put ___ fork down on the table.* [syntax]
- *The woman walked across the street, checking for traffic over ___ shoulder.* [coreference]
- *I went to the ocean to see the fish, turtles, seals, and _____.* [lexical semantics/topic]
- *Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was ___.* [sentiment]
- Iroh went into the kitchen to make some tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the _____. [some reasoning – this is harder]
- I was thinking about the sequence that goes 1, 1, 2, 3, 5, 8, 13, 21, ____ [some basic arithmetic; they don't learn the Fibonacci sequence]
- Models also learn – and can exacerbate racism, sexism, all manner of bad biases.

Sometimes it also memorizes copyrighted material

AI Art Generators Spark Multiple Copyright Lawsuits

Getty and a trio of artists sued AI art generators in separate suits accusing the companies of copyright infringement for pilfering their works.

BY WINSTON CHO JANUARY 17, 2023 4:10PM



BEN STANSALL/AFP VIA GETTY IMAGES



Anthropic fires back at music publishers' AI copyright lawsuit

By Blake Brittain

January 17, 2024 3:30 PM PST · Updated 19 days ago



ARTICLE

Insights from the Pending Copilot Class Action Lawsuit

October 4, 2023

Bloomberg Law

By Daniel R. Mello, Jr.; Jenevieve J. Maerker; Matthew C. Berntsen; Ming-Tao Yang

GitHub Inc. offers a cloud-based platform that is popular among many software programmers for hosting and sharing source code, and collaborating on source code drafting. GitHub's artificial

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

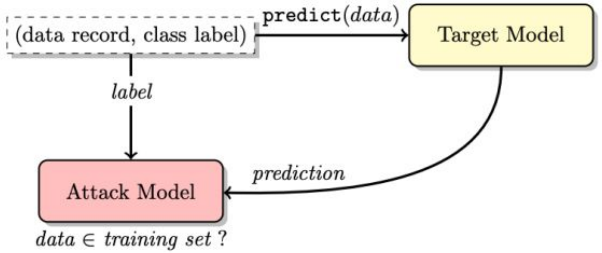
Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

Share free access 1.3K

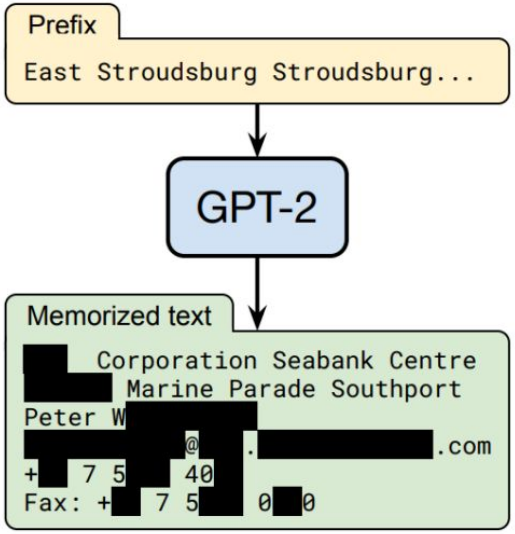


Sometimes it learns some things we don't want..

- *Membership inference* lets you recover parts of the training data



- Sometimes this training data is semi-private material from the web (addresses, emails)



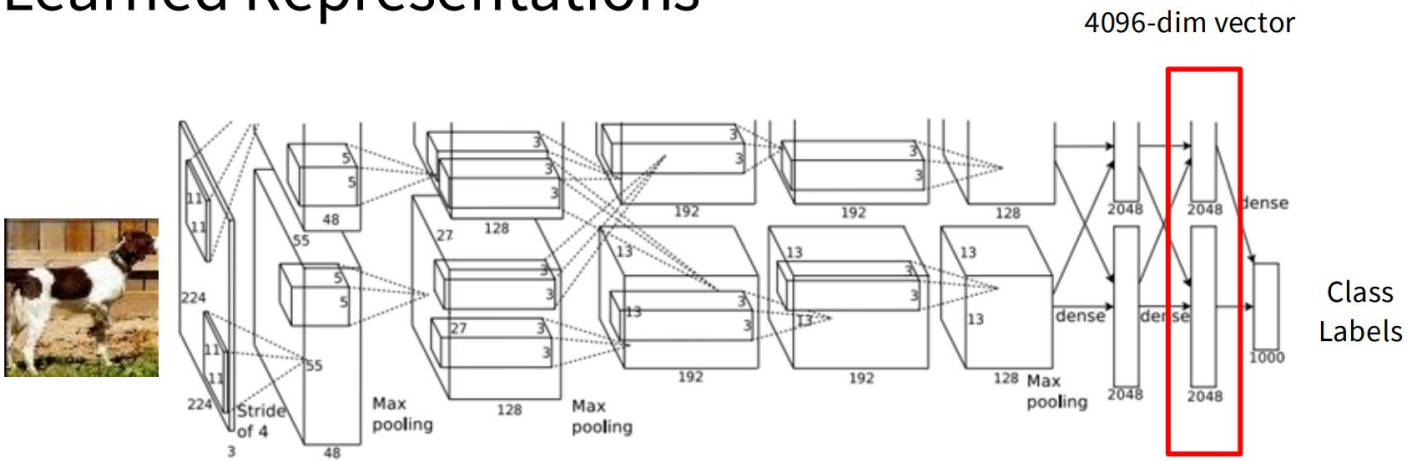
- It learns the prejudices and biases of human beings who write online

Is scaling all u need?



SELF SUPERVISED LEARNING

Learned Representations



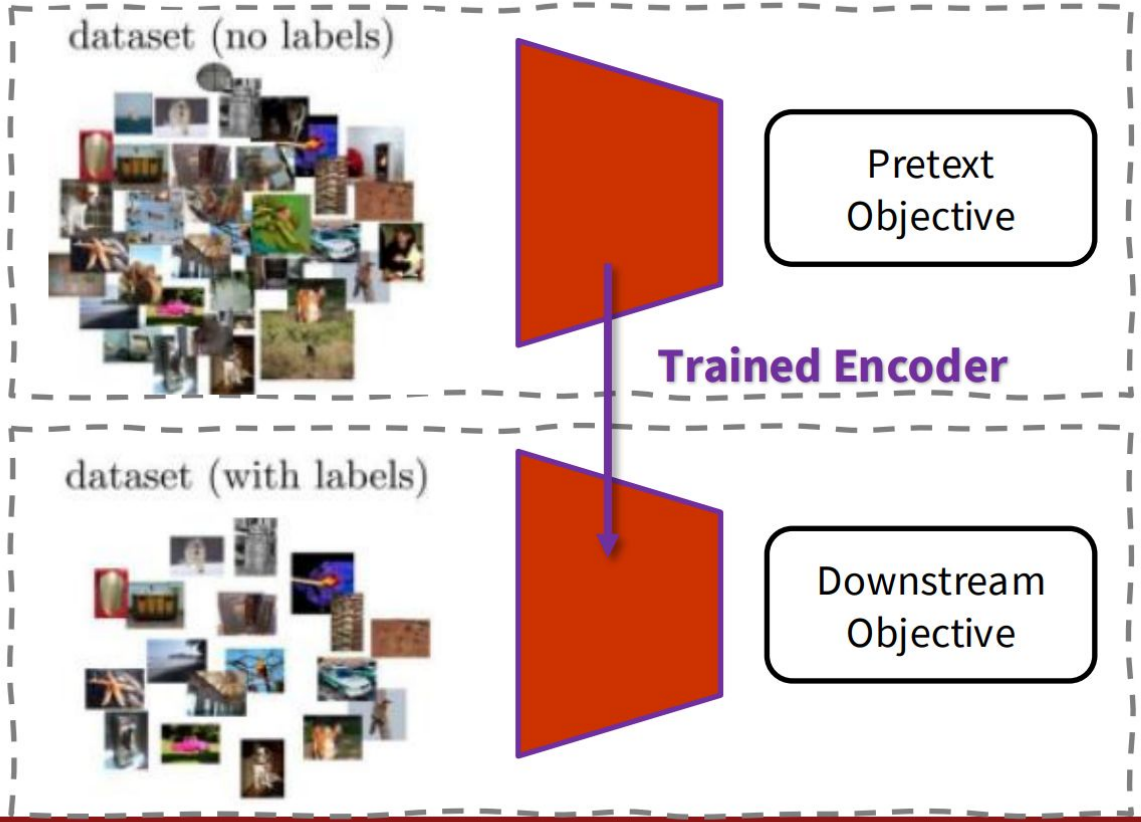
What is the problem with large-scale training?

- **We need a lot of labeled data**

Is there a way we can train neural networks without the need for huge manually labeled datasets?

rizhevsky et al, "ImageNet Classification with Deep Convolutional Neural Networks", NIPS 2012.
gures reproduced with permission.

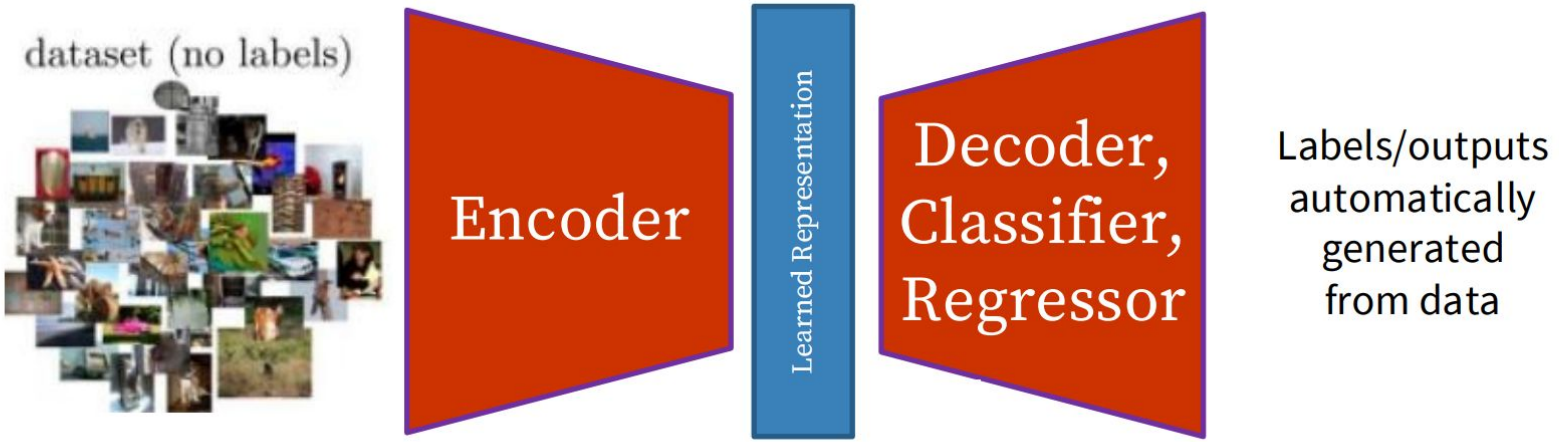
Self-Supervised Learning



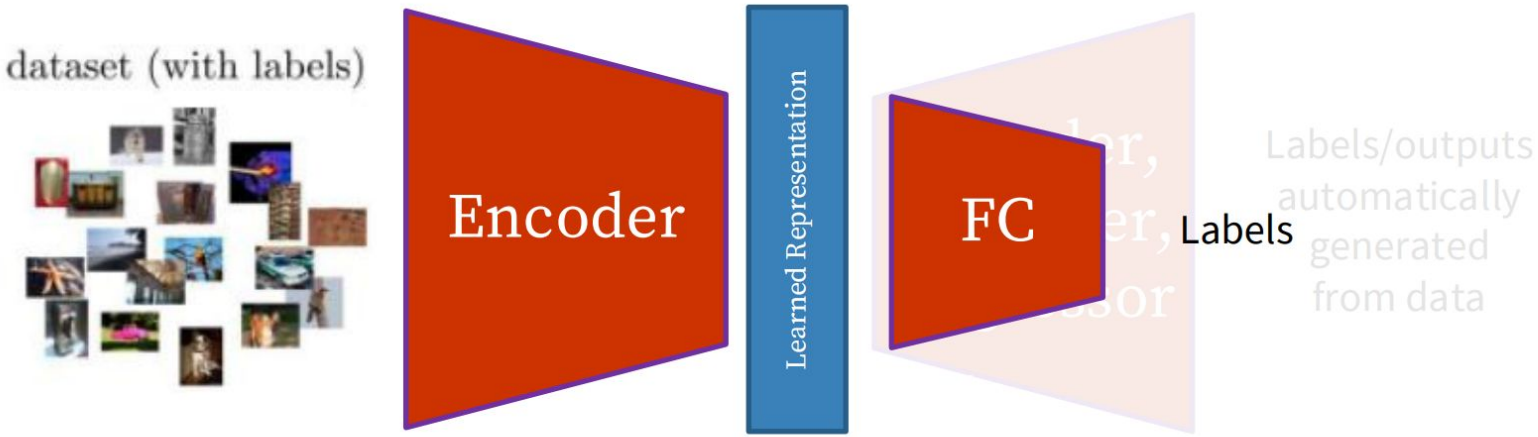
Pretext Task
- Define a task based on the data itself
- No manual annotation
- Could be considered an **unsupervised** task;
- but we learn with supervised learning objectives, e.g., classification or regression.

Downstream Task
- The application you care about
- You do not have large datasets
- The dataset is labeled

Self-Supervised Learning – Pretext Task



Self-Supervised Learning – Downstream Task (supervised)



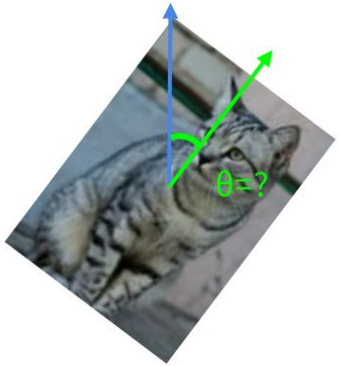
Semi-supervised

Self-supervised pretext tasks

Example: learn to predict image transformations / complete corrupted images



image completion



rotation prediction



“jigsaw puzzle”



colorization

1. Solving the pretext tasks allow the model to learn good features.
2. We can automatically generate labels for the pretext tasks.

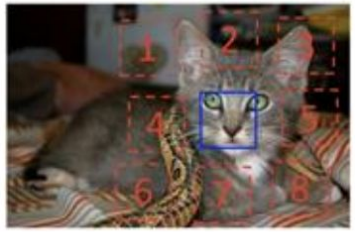
How to evaluate a self-supervised learning method?

- **Pretext Task Performance**
  - Measure how well the model performs on the task it was trained on without labels.
- **Representation Quality**
  - Evaluate the quality of the learned representations
    - *Linear Evaluation Protocol:* Train a linear classifier on the leaerned representations;
    - *Clustering:* Measure clustering performance;
    - *t-SNE:* Visualize the representations to assess their separability.)
- **Robustness and Generalization**
  - Test how well the model generalizes to different datasets and is robust to variations.
- **Computational Efficiency**
  - Assess the efficiency of the method in terms of training time and resource requirements.
- **Transfer Learning and Downstream Task Performance**
  - Assess the utility of the learned representations by transferring them to a downstream supervised task.

Broader picture

Today's lecture

computer vision



Doersch et al., 2015

language modeling

GPT-4 Technical Report

OpenAI*

Abstract

We report the development of GPT-4, a large-scale, multimodal model which can accept image and text inputs and produce text outputs. While less capable than humans in many real-world scenarios, GPT-4 exhibits human-level performance on various professional and academic benchmarks, including passing a simulated bar exam with a score around the top 10% of test takers. GPT-4 is a Transformer-based model pre-trained to predict the next token in a document. The post-training alignment process results in improved performance on measures of factuality and adherence to desired behavior. A core component of this project was developing infrastructure and optimization methods that behave predictably across a wide range of scales. This allowed us to accurately predict some aspects of GPT-4's performance based on models trained with no more than 1/1,000th the compute of GPT-4.

GPT-4 (OpenAI 2023)

speech synthesis



Wavenet (van den Oord et al., 2016)

robot / reinforcement learning



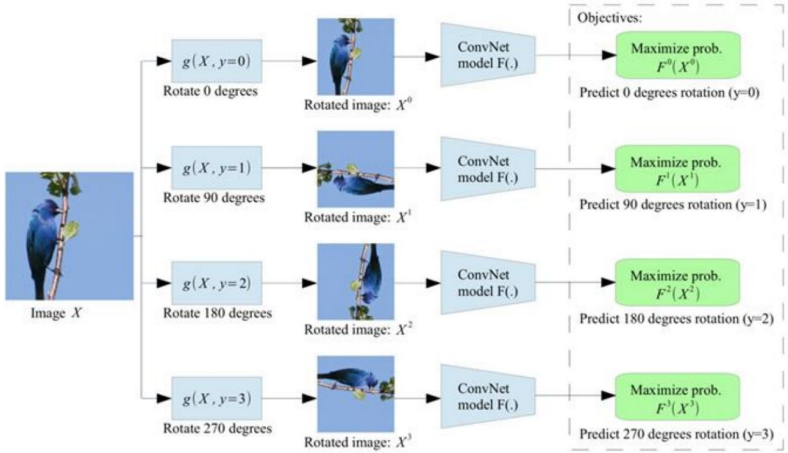
Dense Object Net (Florence and Manuelli et al., 2018)

• • •

Pretext tasks from image transformations

- Rotation, inpainting, rearrangement, coloring
- Reconstruction-based learning (MAE)

Pretext task: predict rotations

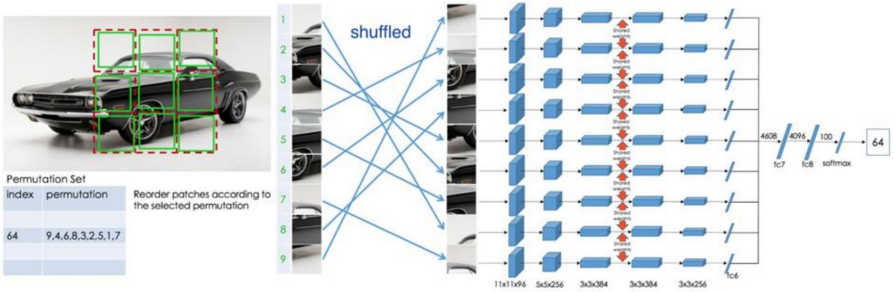


Self-supervised learning by rotating the entire input images.

The model learns to predict which rotation is applied (4-way classification)

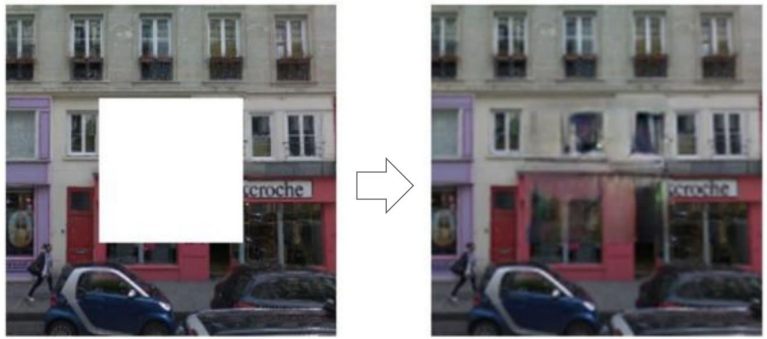
(Image source: Gidaris et al. 2018)

Pretext task: solving “jigsaw puzzles”



(Image source: Noroozi & Favaro, 2016)

Pretext task: predict missing pixels (inpainting)



Context Encoders: Feature Learning by Inpainting (Pathak et al., 2016)

Source: Pathak et al., 2016

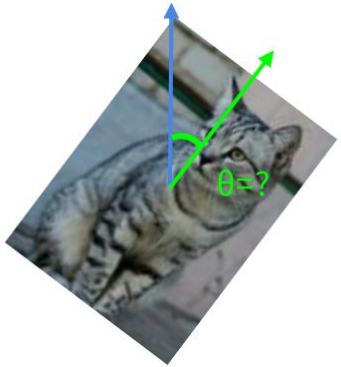
Summary: pretext tasks from image transformations

- Pretext tasks focus on "visual common sense", e.g., predict rotations, inpainting, rearrangement, and colorization.
- The models are forced learn good features about natural images, e.g., semantic representation of an object category, in order to solve the pretext tasks.
- We often do not care about the performance of these pretext tasks, but rather how useful the learned features are for downstream tasks (classification, detection, segmentation).
- Problems: 1) coming up with individual pretext tasks is tedious, and 2) the learned representations may not be general.

Pretext tasks from image transformations



image completion



rotation prediction



"jigsaw puzzle"



colorization

Learned representations may be tied to a specific pretext task!

Can we come up with a more general pretext task?

NEXT LECTURE !!!!!